Matrix Equations

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Fact. *The matrix equation $A\boldsymbol{x} = \boldsymbol{b}$ has a solution if and only if $\boldsymbol{b}$ is a linear combination of the columns of A.*

In particular:

Fact. *Testing whether a vector $\boldsymbol{b}$ is in the span of some collection of vectors is equivalent to asking whether the augmented matrix with those columns is consistent.*

When is Solution Guaranteed?

Fact. *Matrix equation $A\boldsymbol{x} = \boldsymbol{b}$ has a solution for every vector $\boldsymbol{b}$*
$\iff$ *the columns of A span $\mathbb{R}^m$*
$\iff$ *A has a pivot in each row.*

Proof uses the fact that if there is a pivot in each row, then the condition for inconsistent system cannot be satisfied. And if there is not a pivot in each row, then we can choose $\boldsymbol{b}$ where the condition for inconsistent system is true.

Homogenous Systems

Defn. *A* ***homogeneous system*** *is* $A\boldsymbol{x} = \boldsymbol{0}$. *It always has at least the* ***trivial*** solution $\boldsymbol{x} = \boldsymbol{0}$.

Parametric Vector Form of the Solution

ALGOR *The solution to a general linear system can be written in* ***parametric vector form*** *as: one vector plus an arbitrary linear combination of vectors satisfying the corresponding homogeneous system.*

An Example of Parametric Vector Form

For example, we saw earlier following reduced form:

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -3 & -1 \\ 0 & 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

An Example of Parametric Vector Form (cont)

Earlier we gave solution as

$$\begin{aligned} x_1 &= -1 - 2x_2 + 3x_4 \\ x_3 &= 5 - x_4 \end{aligned}$$

Now we add the equations $x_2 = x_2$ and $x_4 = x_4$. This system has solution:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 5 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 3 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Summary

The matrix equation $A\boldsymbol{x} = \boldsymbol{b}$ has a solution if and only if $\boldsymbol{b}$ is a linear combination of the columns of A. This is guaranteed precisely when the columns of A span $\mathbb{R}^m$; equivalently A has a pivot in each row.

A homogeneous system equals **0**. Parametric vector form represents the solution as: one vector plus arbitrary linear combination of vectors satisfying the homogeneous system.